

Evaluating Monetary Policy Counterfactuals Caravello, McKay & Wolf (2026)

Replication using R and C++

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Section 1

Background: Key Concepts

What is New Keynesian Economics?

New Keynesian economics is a school of macroeconomic thought that:

- Builds on Keynesian ideas (prices/wages don't adjust instantly)
- Incorporates *micro-foundations*: derives aggregate behavior from individual optimization
- Assumes *rational expectations*: agents use all available information to forecast
- Features *nominal rigidities*: prices/wages are “sticky” (adjust slowly)

Core insight: Monetary policy has *real effects* because prices don't adjust immediately

Model Abbreviations Explained

Abbreviation	Full Name	Key Feature
RANK	Representative Agent New Keynesian	Single “average” household
HANK	Heterogeneous Agent New Keynesian	Many different households (rich, poor, etc.)
B-RANK	Behavioural RANK	Agents have bounded rationality
B-HANK	Behavioural HANK	Heterogeneous + bounded rationality

- Representative Agent: Assumes economy behaves “as if” there’s one typical household
- Heterogeneous Agents: Different households with different income, wealth, and constraints
- Behavioural: Agents don’t have perfect foresight; they use simpler forecasting rules

Key Technical Terms

IRF (Impulse Response Function): How a variable (e.g., inflation) responds over time to a one-time shock (e.g., interest rate change)

Wold Representation/Response: A way to express any stationary time series as an infinite weighted sum of past “innovations” (unpredictable shocks) — captures the *average* business cycle dynamics

Policy Causal Effects (Θ''): The true effect of changing the policy rate on macroeconomic outcomes, isolated from other factors

BVAR (Bayesian VAR): Vector Autoregression estimated with Bayesian methods (prior beliefs + data)

DSGE: Dynamic Stochastic General Equilibrium model — a fully specified structural model

Why This Paper Matters

Traditional approach to policy counterfactuals:

- 1 Build a full structural DSGE model
- 2 Change the policy rule
- 3 Re-solve the entire model

Problem: Results depend heavily on model assumptions we can't test

This paper's insight:

- Many counterfactuals only need *reduced-form evidence* (what we observe)
- Structural models are only needed to *extrapolate* beyond observed policy variation
- When extrapolation is needed, behavioural assumptions matter more than heterogeneity

Section 2

Main Findings

The Central Question

How do we evaluate monetary policy counterfactuals?

Counterfactual: “What would have happened if policy had been different?”

- Traditional approach: Build micro-founded DSGE model, change policy, re-solve
- Semi-structural approach: Directly leverage empirical evidence from historical policy variation

This paper bridges both traditions:

Use as much reduced-form empirical evidence as possible, supplement with minimal structure necessary

Historical Context: Saltwater vs Freshwater Economics

School	Universities	Core View
Saltwater	MIT, Harvard, Berkeley, Princeton	Markets fail; policy intervention valuable
Freshwater	Chicago, Minnesota, Rochester	Markets efficient; policy often harmful

Relevance to this paper:

- *Freshwater* tradition: Rational expectations, micro-founded DSGE models, structural approach
- *Saltwater* tradition: More open to reduced-form evidence, policy pragmatism

The debate:

- Can we trust reduced-form evidence, or do we *need* structural models to understand policy effects?

This paper's answer:

- Both have a role
- Use reduced-form evidence when possible (saltwater pragmatism), but structural models fill gaps where data is silent (freshwater discipline)

Key Result: Sufficient Statistics

To evaluate monetary policy counterfactuals, you only need two objects:

① Policy Causal Effects Θ^ν

- IRFs of all possible policy rate changes on macro outcomes
- Effects across the entire yield curve (short-end to long-end)
- *What happens to inflation/output when we change interest rates?*

② Reduced-Form Projections

- Wold impulse responses (the average business cycle)
- Forecasts (where the economy is heading without intervention)

⇒ You do not need the full structural model, shock processes, or the policy rule itself!

What Empirical Evidence Tells Us

The constraint: Existing monetary policy shock literature primarily identifies effects of *transitory* rate changes (short end of yield curve)

Note: “Short end” = immediate/near-term rate changes; “Long end” = persistent/future rate changes

- Romer-Romer (2004): Short-lived policy rate changes
- Aruoba-Drechsel (2024): Somewhat more persistent
- Gertler-Karadi (2015), Miranda-Agrippino-Ricco (2023): Similar

Missing piece: Causal effects of *delayed or persistent* rate changes (what happens when central banks commit to *future* rate paths)

⇒ Need structural models to extrapolate beyond observed data variation

The Extrapolation Question

Four structural models for extrapolation:

Model	Agents	Expectations	Description
RANK	Representative	Rational	Single “average” agent, perfect foresight
HANK	Heterogeneous	Rational	Many agents (rich/poor), perfect foresight
B-RANK	Representative	Behavioural	Single agent, bounded rationality
B-HANK	Heterogeneous	Behavioural	Many agents, bounded rationality

Key finding: Once matched to short-end IRFs:

Why HANK $\approx$ RANK (When Matched)

Intuition:

- 1 Market incompleteness can dampen or amplify monetary transmission
- 2 But: matching to *identical short-end effects* means heterogeneity channels largely offset
- 3 Result: “as-if” aggregation (echoes Werning, 2015)

Plain language: When we force both models to match the same observed data (how economies respond to rate changes), the extra complexity of modeling different household types doesn't change the aggregate predictions much.

Implication: For aggregate counterfactuals, whether we use RANK or HANK is largely immaterial

What matters: Strength of behavioural frictions (how people form expectations about the future)

Section 3

Methodology

Bayesian VAR Estimation

VAR (Vector Autoregression): A statistical model where each variable depends on its own past values and past values of all other variables.

Estimate reduced-form dynamics using BVAR with flat prior:

$$Y_t = B_1 Y_{t-1} + \dots + B_p Y_{t-p} + u_t$$

Variables included:

- Unemployment, Output, Investment, Consumption
- Hours, TFP, Labor Productivity, Labor Share
- Inflation, Federal Funds Rate

Sample: 1960-1 to 2019-4, quarterly data

Computing Wold IRFs

Wold representation: $y_t = \sum_{\ell=0}^{\infty} \Psi_{\ell} u_{t-\ell}$

Intuition: Any stationary series can be written as a weighted sum of past “surprises” (u_t). The weights (Ψ_{ℓ}) tell us how past shocks propagate over time.

Implementation:

- 1 Decompose $\Sigma = LL'$ (Cholesky)
- 2 Compute Wold coefficients recursively from companion form
- 3 Apply rotation: $\text{IRF}_h = \Theta_h \cdot L$

I was able to use R and armadillo4r to recreate the same figures as the article.

- $\Rightarrow$ advantage: free as in free of charge replication, open source and cross-platform independent validation
- $\Rightarrow$ disadvantage: not free in time, Armadillo requires some C++ knowledge (not as user-friendly as MATLAB)

See <https://github.com/pachadotdev/ecod024/tree/main/cmw26> for the replication code.

Counterfactual Computation

Question: “What would have happened under a different policy rule?”

Given policy rule $A_\pi \pi + A_y y + A_i i = \omega$:

(Example: Taylor rule weights inflation and output gap)

Find optimal shock sequence:

$$\nu^* = -(M' M)^{-1} M' \cdot \text{target}$$

where $M = A_\pi \Pi_m + A_y Y_m + A_i I_m$

Counterfactual outcomes:

$$\tilde{\pi} = \pi + \Pi_m \nu^*, \quad \tilde{y} = y + Y_m \nu^*, \quad \tilde{i} = i + I_m \nu^*$$

Optimal Policy Computation

Dual mandate: Central banks (e.g., the Fed) aim to balance two goals:

- 1 Price stability (low inflation: π)
- 2 Maximum employment (output near potential: y)

Loss function captures this trade-off:

$$\mathcal{L} = \sum_{t=0}^T \beta^t [\lambda_{\pi} \pi_t^2 + \lambda_y y_t^2 + \lambda_{\Delta i} (i_t - i_{t-1})^2]$$

Third term: penalize rapid rate changes for policy stability

Solve for optimal policy path via quadratic programming:

$$\min_{\nu} \frac{1}{2} \nu' H \nu + f' \nu$$

Section 4

Replication

Computational Complexity

Function	Complexity	Description
bvar_fn	$O(D \cdot n^3)$	D posterior draws, n^3 for Wishart sampling
var_irf	$O(H \cdot n^2 \cdot p)$	H horizons, p lags, matrix products
cnfctl_fn	$O(T^3)$	Solves $(M' M)^{-1} M'$ with $T \times T$ matrices
optpol_fn	$O(T^3)$	QP solve via $H^{-1} g$ with $T \times T$ Hessian
freq_var_fn	$O(n_\omega \cdot H \cdot n^2)$	Spectral integration over frequency grid

Bottleneck: Counterfactual Loop

The main computational burden is the counterfactual loop:

$$\text{Total operations} \approx D \times M \times T^3$$

where $D = 1000$ draws, $M = 4$ models, $T = 200$ horizons.

For each draw d and model m :

- 1 Extract $\Pi_m^{(d)}, Y_m^{(d)}, I_m^{(d)}$ from sufficient statistics
- 2 Solve $\nu^* = -(M' M)^{-1} M' \cdot \text{target} \rightarrow O(T^3)$
- 3 Compute counterfactual paths $\rightarrow O(T^2)$

$\Rightarrow$ Historical evolution (Figure 7) runs this $\times 20$ forecast dates

$\Rightarrow$ Replication time: ~ 1 hour on a standard laptop

Results: Figures 3-5

Figure 3-2: Model IRFs – Romer–Romer Shock

Shaded band = Empirical VAR (68% CI)

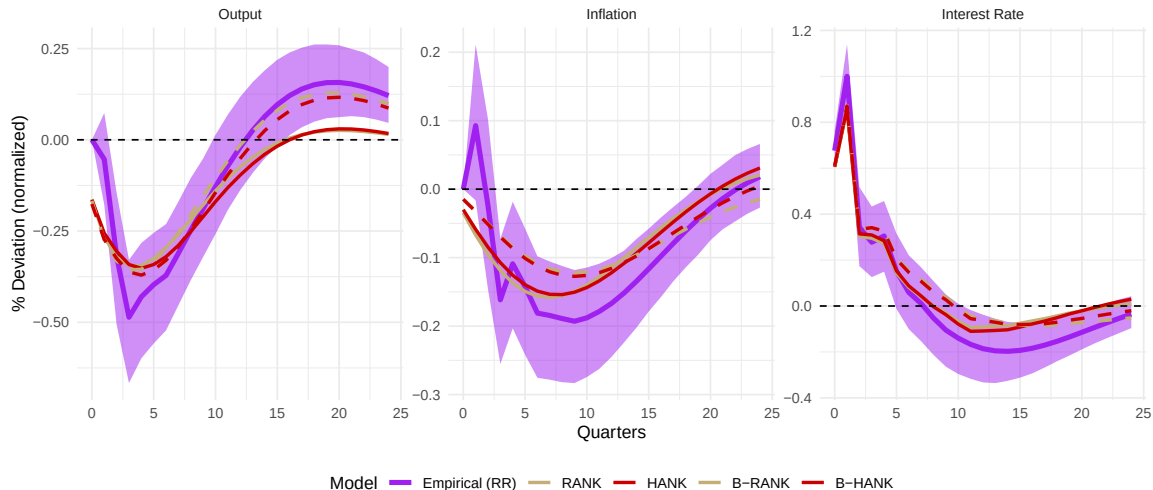


Figure 3–1: Model IRFs – Aruoba–Drehssel Shock
Shaded band = Empirical VAR (68% CI)

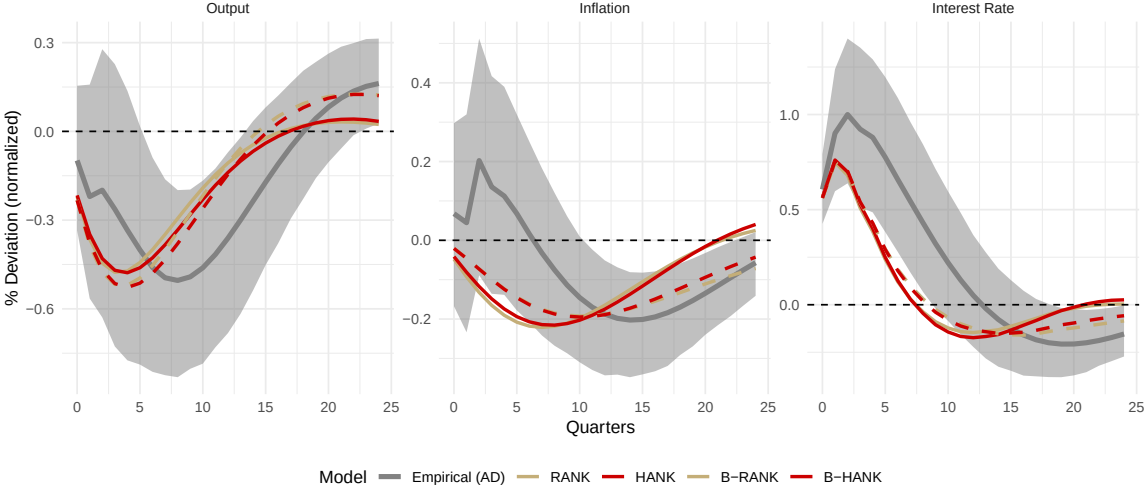


Figure 4: News Shock IRFs – RANK vs HANK

Response to monetary policy shock anticipated 16 quarters ahead (68% bands)

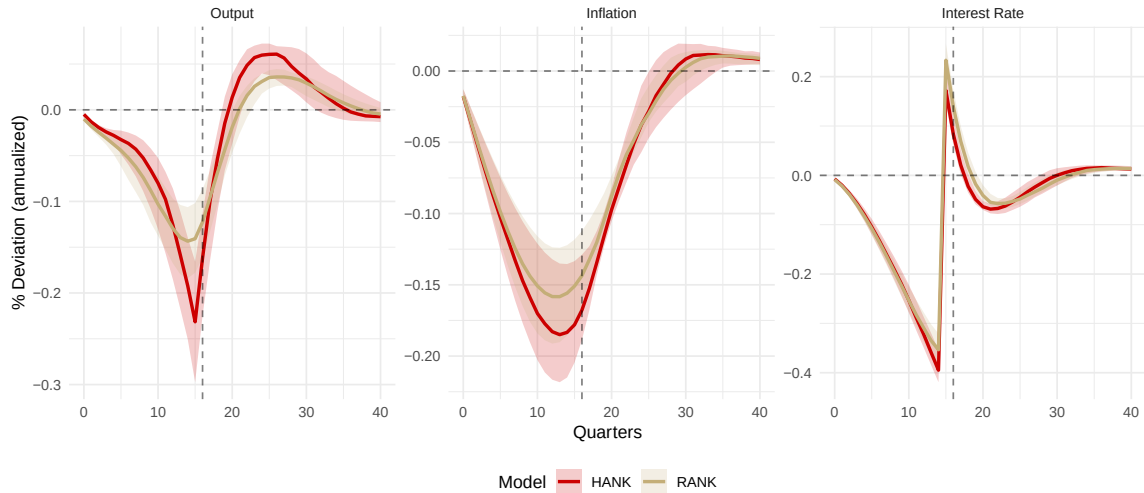


Figure 5: News Shock IRFs – RANK vs B-RANK

Response to monetary policy shock anticipated 16 quarters ahead (68% bands)

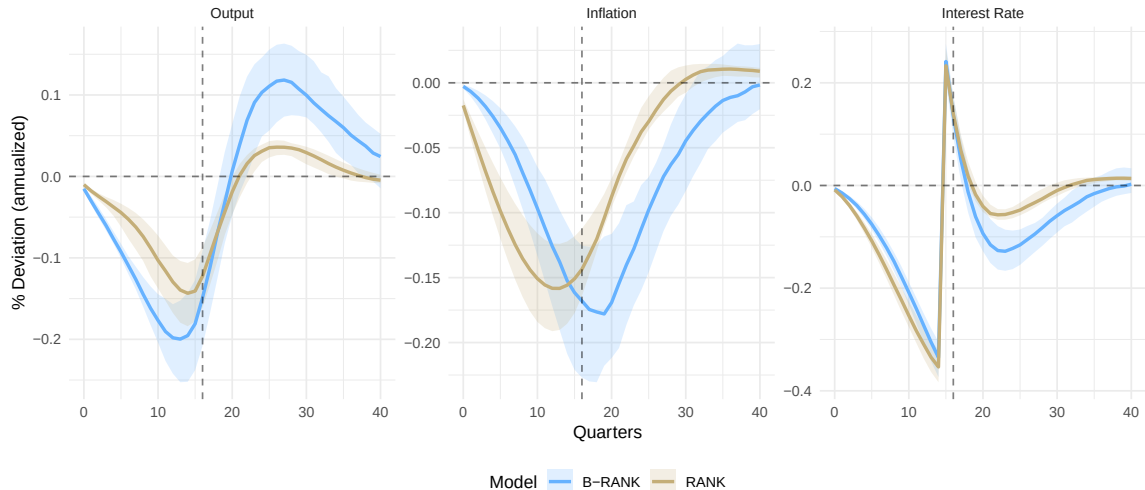
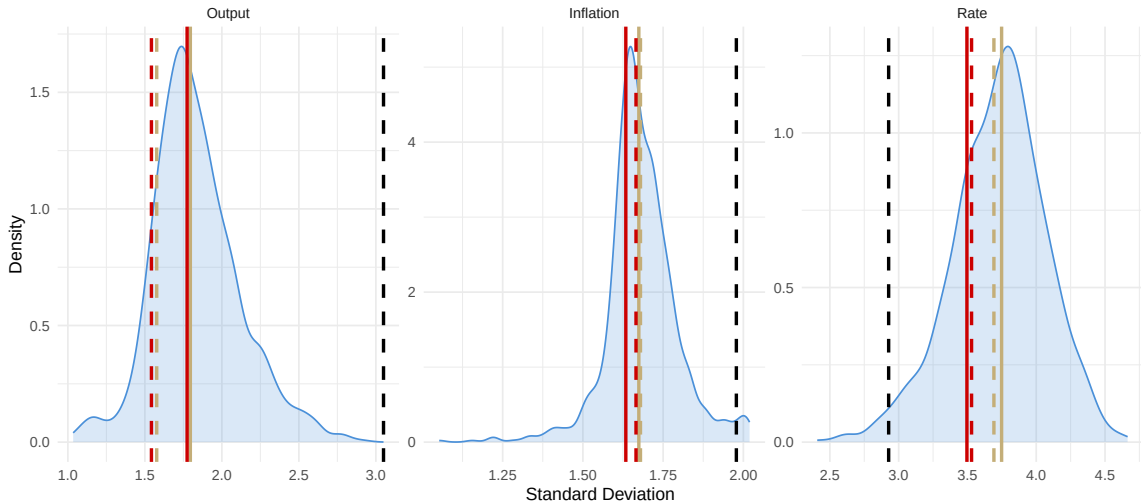


Table 5.1: Counterfactual Second Moments

Approach	Model	Inflation	Output	Interest Rate
Actual		1.978	3.050	2.927
Hybrid	RANK	1.674 (1.607, 1.772)	1.795 (1.592, 2.091)	3.748 (3.408, 4.048)
Hybrid	HANK	1.634 (1.558, 1.733)	1.774 (1.554, 2.101)	3.496 (3.203, 3.758)
Hybrid	B-RANK	1.679 (1.582, 1.858)	1.577 (1.340, 1.831)	3.692 (3.328, 4.117)
Hybrid	B-HANK	1.666 (1.595, 1.844)	1.543 (1.297, 1.811)	3.530 (3.214, 3.871)
Semi-structural		1.795 (1.628, 1.913)	2.153 (1.913, 2.422)	2.841 (2.512, 3.183)

Figure D.1: Counterfactual Unconditional Volatilities

Black = observed policy; Blue = RANK posterior density; Solid Beige = RANK; Dashed Beige = B-RANK; Solid Red = HANK; Dashed Red =



Figures 6-8

Figure 6: Counterfactual Response to Main Business Cycle Shock

Data (black dotted); Hybrid (blue band/line); Semi-structural (orange band/line)

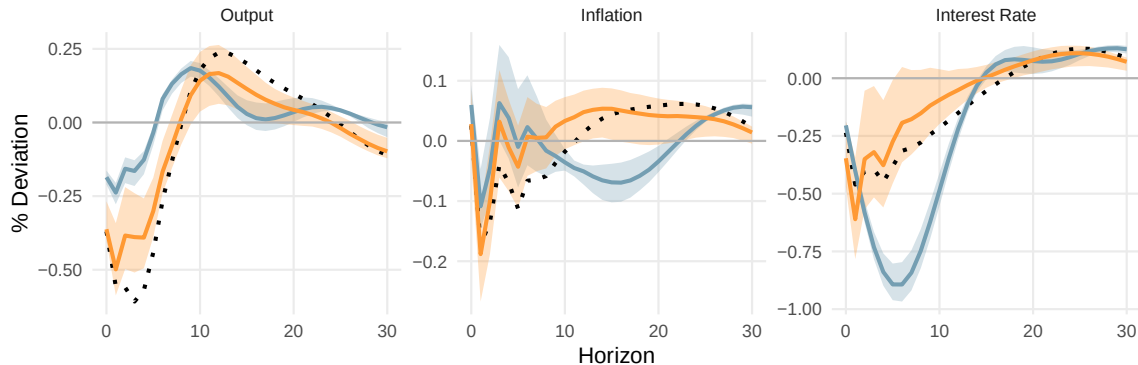


Figure 7: Historical Evolution Under Counterfactual Policy (Great Recession) – RANK

Black: data; Beige: RANK (dashed=behavioral); Blue: RANK bands

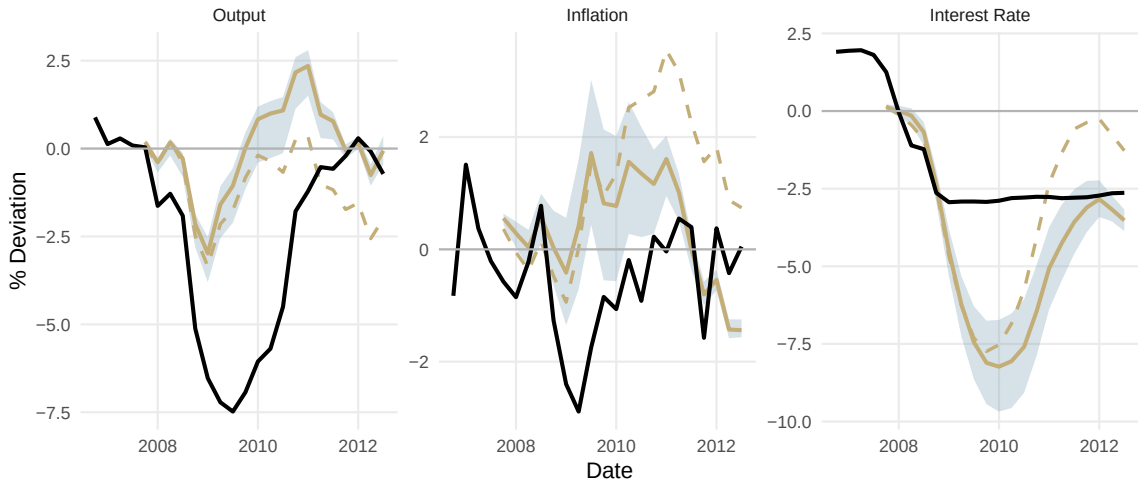


Figure 7: Historical Evolution Under Counterfactual Policy (Great Recession) – HANK

Black: data; Red: HANK (dashed=behavioral)

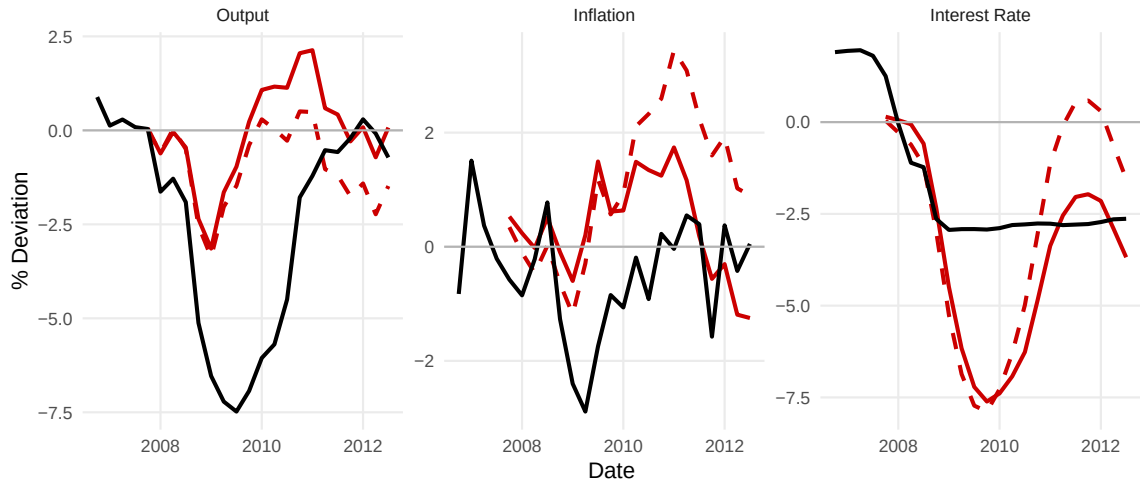


Figure 7: Historical Evolution Under Counterfactual Policy (Great Recession) – Semi-Stru
Black: data; Orange: semi-structural

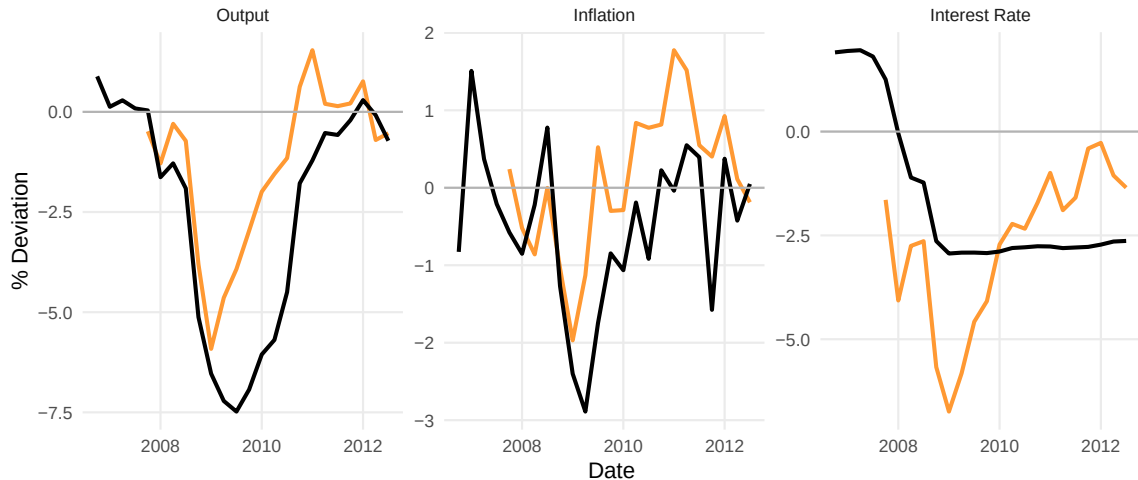


Figure 8: Historical Scenario (Post-COVID) – RANK vs HANK

Black: data | Grey dashed: VAR forecast | Beige: RANK | Red: HANK

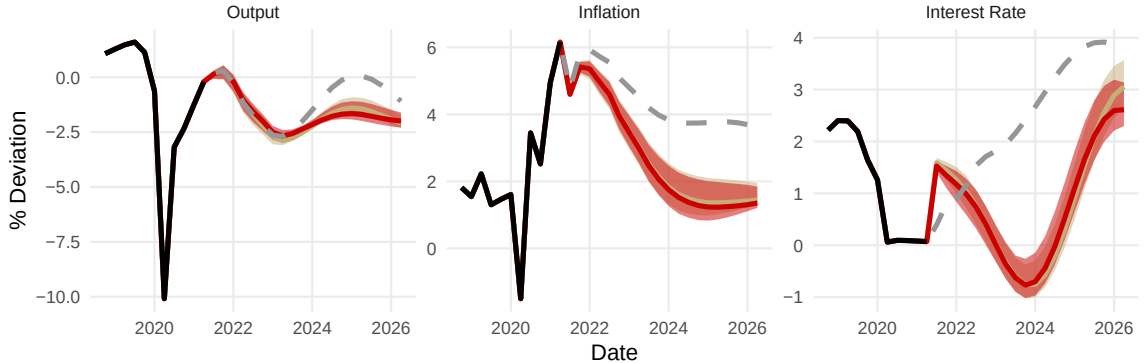


Figure 8: Historical Scenario (Post-COVID) – RANK vs B-RANK

Black: data | Grey dashed: VAR forecast | Beige: RANK | Blue: B-RANK

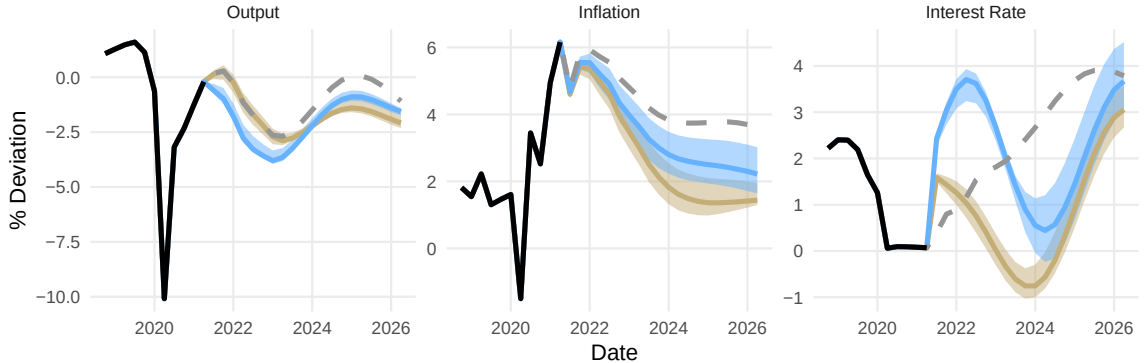
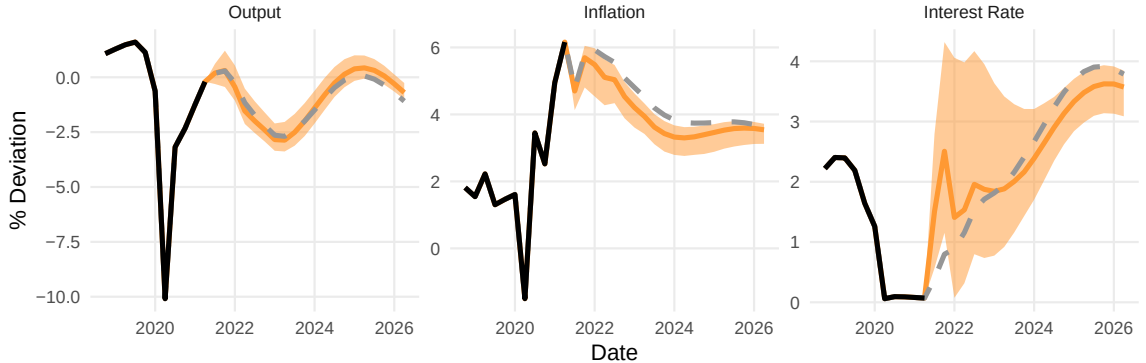


Figure 8: Historical Scenario (Post-COVID) – Semi-structural

Black: data | Grey dashed: VAR forecast | Orange: semi-structural



Section 5

Key Takeaways

Summary of Findings

- ① Semi-structural approach often suffices $\implies$ When counterfactual is about near-term policy, model extrapolation adds little
- ② $RANK \approx HANK$ for aggregates $\implies$ Once matched to short-end IRFs, household heterogeneity doesn't matter much for aggregate outcomes
- ③ Behavioural frictions are crucial
 - $\implies$ *Forward guidance* effectiveness depends critically on expectation formation
 - $\implies$ B-RANK/B-HANK predict much weaker effects of anticipated policy
 - (Forward guidance = central bank announcing future policy intentions)
- ④ Great Recession: Dual mandate central bank would have preferred deeply negative rate
- ⑤ Post-COVID: Standard models suggest forward guidance could stabilize inflation; behavioural models disagree

Implications for Policy Analysis

Practical guidance for counterfactual evaluation:

- Check if contemplated policy changes are mainly about *near-term* rates
 - If yes $\implies$ semi-structural approach likely sufficient
 - If no $\implies$ need structural model for extrapolation
- For extrapolation, focus on:
 - Matching short-end policy effects (discipline the model)
 - Behavioural assumptions about expectations (key for long-end effects)
 - *Not* on RANK vs HANK distinction